

Paper 1 PD samples

Sample C

Question 1

Hormones are examples of chemical messengers, and some hormones may affect the brain and nervous system. Cortisol is a hormone produced in the adrenal glands and is thought to play a role in stress. High levels of cortisol can negatively affect memory retrieval because it can lead to hippocampal impairment.

Newcomer's research investigated if high levels of cortisol would harm verbal declarative memory. The participants were given a tablet containing high, medium or no cortisol over 4 days. They were then asked to listen to and later recall parts of a prose paragraph. The results indicated that high cortisol levels impaired performance in the memory task for those who received the highest dose compared to the other two groups.

The findings show that high levels of cortisol interfered with the effective recall of information. This provides evidence that the chemical messenger cortisol can interfere with normal brain function and lead to memory impairment.

Question 2

The term "anchoring bias" is a cognitive bias in which people make decisions or conclusions based unduly on the first piece of information they are given, or the "anchor."

Anchoring bias significantly impacts prejudice by shaping individuals' perceptions and judgments of others based on initial, often arbitrary, information or stereotypes.

Even if the first piece of information is false or irrelevant, it still affects subsequent decisions. The dual-process theory, which argues that human thinking comprises of both slower, more deliberate processes and automatic, intuitive processes, provides an explanation for anchoring bias.

People rely on automatic processing while making decisions or passing judgment. This includes a tendency to use anchors as points of reference without questioning their applicability or correctness. The concept of heuristics—mental shortcuts humans employ to quickly form opinions and decisions—is related to anchoring bias. Social cognition, decision-making, and problem-solving are all affected by anchoring bias. For example, anchoring bias in social cognition can affect how impressions and stereotypes are formed because people tend to use their first impressions or preconceptions as anchors when making judgments about other people. People's interpretation of later information is influenced by this bias, which makes them selectively focus on details that support their preconceptions and ignore opposing data. Stereotypes regarding particular social groupings are examples of anchors: they act as points of reference that skew people's perceptions and decision-making. As a result, anchoring bias contributes to discrimination and inequality in society by promoting and maintaining biased attitudes and behaviors.

Thus, the anchoring bias can also contribute to confirmation bias, the tendency to seek out and interpret information in a way that confirms one's pre-existing beliefs or stereotypes.

To overcome the impact of anchoring bias on prejudice, it's important for individuals to engage in critical thinking and reflection, question their initial judgments and stereotypes, and actively seek out diverse perspectives and information.

Section B

Question 1

Models can play an extensive role in studying cognitive processes and help predict and explain certain cognitive phenomena. This study allows the researchers to test the model and hypotheses they have created for imagination and understand whether it is their apt representation of the cognitive process.

This is similar to Conrad and Hall's experiment to understand the working model of memory. The working memory model was a multi component system and zoomed into understanding the structure of the short-term memory store, similarly, the current study suggests different stages and components to imagination. Imagination may be considered a part of schema theory which is largely based on mental organisation. This is why models help us study, cognitive processes because it can organise different aspects and components that may play a role and simplify complex, cognitive behaviours and processes. Models, such as these help provide a basis to develop theories on the cognitive process they investigate. With respect to this study, the researchers tested their model using children who proved to be cognitively aligned with the predicted basis of the process of imagination. Further studies should be conducted with other models as well to finalise the explanation and theory of imagination. The essential role such models can play in studying cognition include allowing researchers to test their hypotheses, explain it conceptually, and in an organised manner, represent it in a simplified manner, predict future theories and explanations, and provide a basis for further development.

Question 2

Social learning theory assumes that humans learn behavior such as empathy or prosocial behaviour through observational learning. In other words, people can learn by watching models and imitating their behavior.

When we first observe a model, attention must be paid to learn the behavior. Factors influencing attention involve attractiveness of the role model, authority, and desirability. One must also be able to remember the behavior observed to produce it immediately later. This is called retention. Furthermore, observers must want to replicate the behavior observed. This will depend on if they understand the potential outcome of the behavior if they repeat it (if it will be positive or negative). In addition, to carry out the behavior they must be physically or mentally competent to reproduce it. There must also be a certain level of self-efficacy. Someone who has high self-efficacy believes that he/she can master challenging problems, making someone more likely to attempt to imitate the behavior of the model. Another factor which is

involved in social learning is consistency. If the model behaves in a way that is consistent across situations there is a higher chance that their behaviour will be imitated. People also have the tendency to imitate models who are like them (e.g. same age or gender), this is referred to as identification. Furthermore, liking the model also determines if the behavior will be observed and imitated or not. For instance, if the model is warm and friendly, it is more likely that they will be imitated, whereas if they are cold and uncaring, it is less likely that they will be imitated. Lastly, it is suggested that if a model is observed where it is being awarded for a behavior, there are more chances that it will be imitated in comparison to if the model was punished. This is called vicarious reinforcement.

Prosocial behaviour and/or empathy can be socially learnt. A strategy that could be based on the principles of social learning theory to promote prosocial behavior in school children is if members of the school community are publicly praised or rewarded when they demonstrate examples of good citizenship and empathy. When young school children observe older children who are positive role models and who are actively involved in volunteerism, charitable events and mentoring programmes, they may identify with them and see their behaviour as desirable and worthy of imitation as they may be seen as attractive role models. When their own classmates are praised for consistently showing inclusiveness and kindness, they may also identify with them and imitate their behaviour as they are warm and friendly, and the same age so are relevant role models. If members of the school community are praised in school assemblies or receive good citizenship awards, then their behavior is seen as desirable as it has a positive potential outcome. In this way, the strategy can lead to vicarious reinforcement and promote empathy and prosocial behaviour.

Section C

Major Depressive Disorder or depression is a common psychological disorder characterised by low mood, negative thoughts and feelings of hopelessness. One perspective that explains depression is the biological approach and specifically the genetic explanation. Researchers have studied the role of genetic inheritance in the development of depression. but psychologists in the field of health and wellness now know that there are also environmental factors such as our upbringing, stress levels or socioeconomic background which can also influence behaviour. Therefore, it is unlikely that one perspective alone can effectively explain the disorder of depression.

The genetic explanation for depression suggests that the genes we inherit from our parents may predispose us towards developing disorders like depression as these genes can influence physiological processes that contribute to such behaviours. The diathesis-stress model however, suggests that genes alone are not responsible for depression as such predispositions are not expressed without the appropriate environmental stimuli. This model proposes that an individual who inherits specific genes

and also experiences a stressful environment which can activate those genes, is more likely to develop depression than an individual who has not inherited the same genes. What is inherited is a genetic vulnerability to depression rather than the disorder itself.

Twin studies are often used to study the genetic basis of depression and are based on the assumption that if concordance rates for the disorder are significantly higher for monozygotic twins than for dizygotic twins or siblings, then there is a genetic component to depression. Kendler et al in 2006 carried out a twin study to determine the role that genetics plays in major depressive disorder. The sample consisted of approximately 15,000 twin pairs from the national Swedish Twin Registry. The researchers used a team of trained interviewers to carry out telephone interviews which assessed the occurrence of lifetime major depression by using modified DSM-IV criteria. The results showed that the heritability of MDD was 0.38 and that concordance rates were highest in female MZ twins with a rate of 0.44 compared to 0.31 for MZ males. The rates for DZ twins were fairly comparable. The findings suggest that heritability of MDD is greater in females than males and that some genetic risks for MDD may be specific to females.

This is a correlational study so we cannot assume a cause-and-effect relationship between genetic inheritance and development of depression. Furthermore, no official diagnosis of depression was made by the researchers and the self-reported depressive symptoms may not be reliable. However, the large sample size may override these limitations. Although the study provides evidence that genes play a role in the development of depression, it is important to consider that the concordance rates, even for MZ twins, were well below 100%. There are clearly other factors involved in depression apart from genetics and this supports the claim that there may be more than one perspective to explain this complex human behaviour, not just the one implied by the results of the study.

Another way in which it can be seen how genes play a role in human behaviour is by looking at the 5-HTT gene and its role in the development of MDD. This is relevant to the diathesis-stress model which predicts that an individual's reaction to stressful life events depends on their genetic make-up. If an individual has a specific genotype, then interaction with the environment may cause these genes to be expressed. The serotonin transporter gene 5-HTT is involved in the reuptake of serotonin in brain synapses and may have a gene-environment interaction which can be explored.

Caspi et al investigated a sample of around 850 New Zealand 26-year olds. They were divided into three groups. Group 1 had two short alleles, group 2 had two long, and group 3 had one short and one long allele. The participants were asked to fill in a "stressful life events" questionnaire on

the frequency of 14 different events including financial, employment, health and relationship stressors between the ages of 21 to 26. They were also assessed for symptoms of depression.

It was seen that those who had inherited one or more short variants of the allele had more symptoms of depression in response to these stressful life events. Just inheriting the short variant was not sufficient to lead to depression but the genes interaction with stressful life events increased the participants chances of developing MDD. Therefore, genetic inheritance was seen to play a partial role in the development, however, it relied on the gene-environment interaction which increased the possibilities of its expression.

This study was again correlational which means that it cannot be concluded that the specific allele of the 5-HTT gene causes the development of MDD. It may be the salience of the stressful life events which plays a role in depression, and those who recalled them more easily may be those who already have a tendency toward depression, making the findings of the study unreliable. It is important to consider that this study acknowledges the interaction between both biological and environmental factors in depression, so it supports a more holistic approach and is not reductionist as it does not just focus on the biological perspective.

The evidence above shows that there is some genetic influence in the development of depression, but the diathesis-stress model would suggest that both genes and the environment play a role in its development. In fact, modern research in epigenetics considers that there is a bidirectional relationship between our genetic inheritance and environmental factors as our genes can influence the environments we seek and how we react to them. The environments we function in can also affect how our genes are expressed. For example, in Brown & Harris's study on women and depression, they found that women who had fewer protective social factors such as spousal or community support were more likely to get depression. It is most likely, however, because of gene-environment interaction that the women developed the disorder.

Therefore, it is important to consider that just one perspective, in this case, the biological perspective, limits our understanding of complex conditions such as depression. To avoid an over-simplistic and reductionist conclusion about the role of genes in depression, a deeper understanding of the interplay between genes and the environment in which such genes are expressed is required.